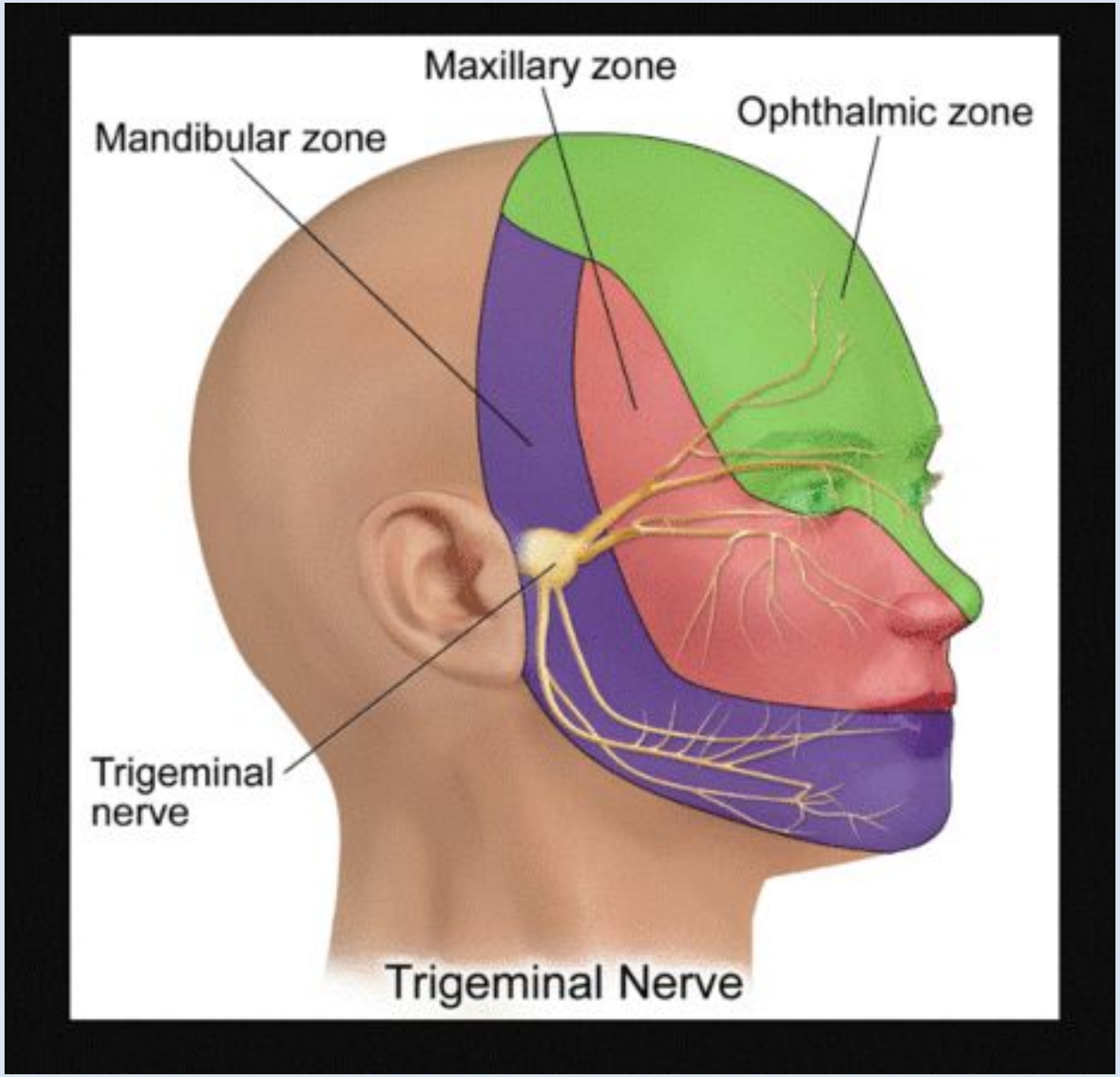
MOTOR FUNCTIONS OF TRIGEMINAL NERVE

DR SHAHNEELA SIRAJ

DIKIOHS,DUHS.

LEARNING OBJECTIVES

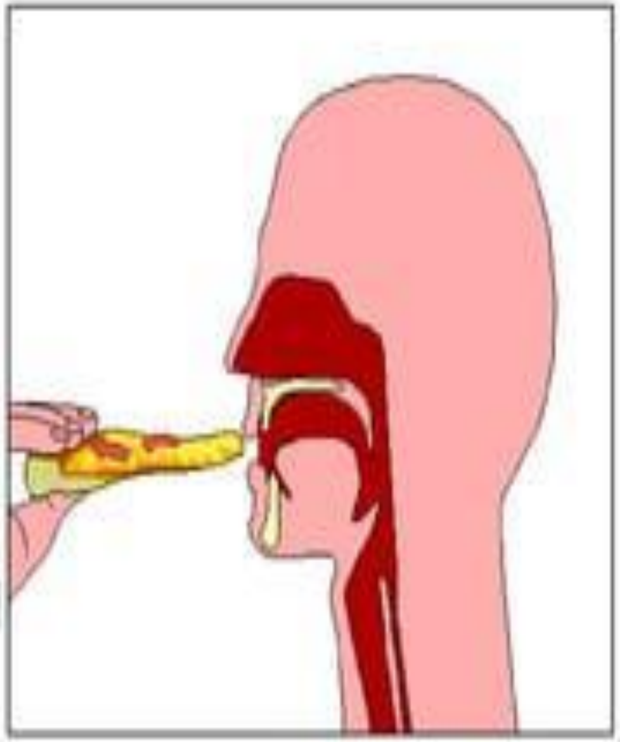
- At the end of the lecture, students should be able to;
- Identify muscles of mastication
- Relate nerve supply to movements
- Describe phases of mastication
- Explain coordination of jaw movements
- Relate mastication to oral functions


Maxillary zone
Ophthalmic zone
Mandibular zone
Trigeminal
nerve
Trigeminal Nerve

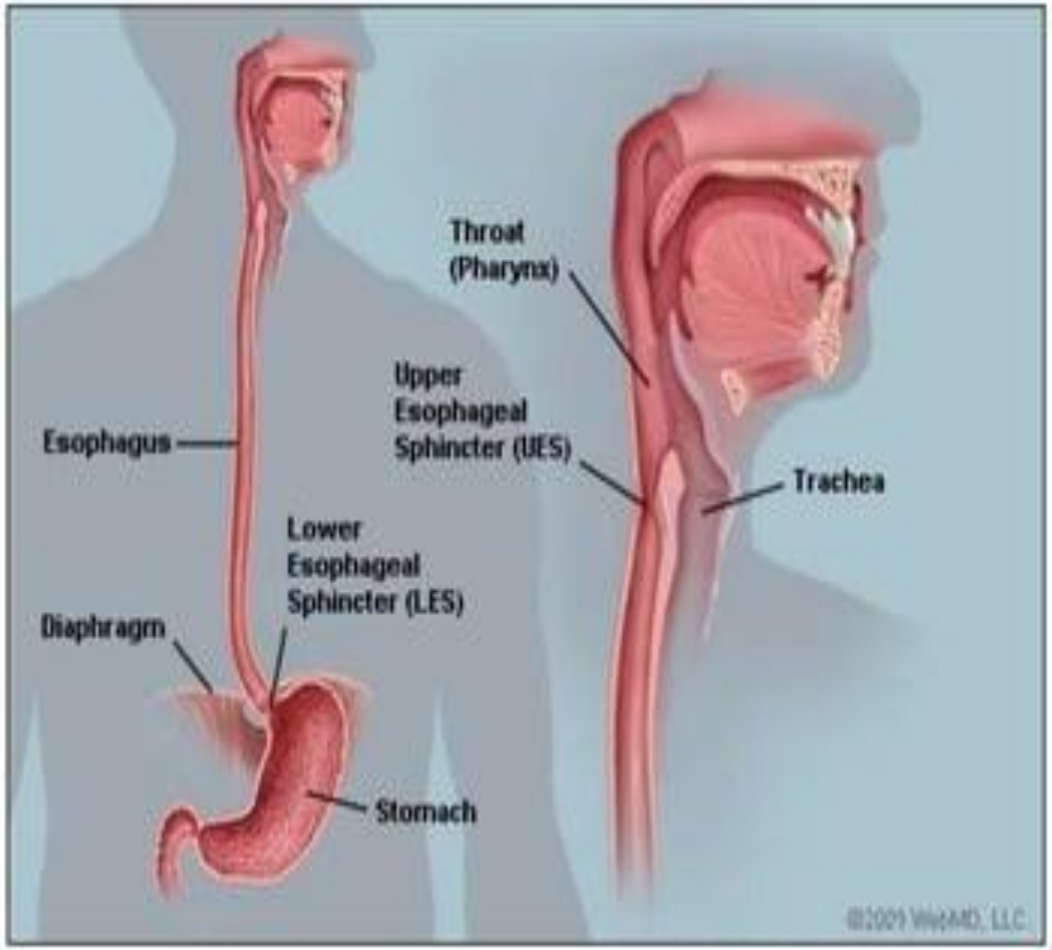
- **The motor root of the mandibular nerve innervates the four muscles of mastication**
- Masseter,
- Temporalis muscle,
- The lateral pterygoids and
- The medial pterygoids.
- The muscles of mastication produce elevation, depression, protrusion, retraction, and the side-to-side movements of the mandible.
- Additionally, the motor component of the V3 also innervates the tensor veli palatini, the mylohyoid, the tensor tympani, and the anterior portion of the digastric muscle

INTRODUCTION

MASTICATION is the process by which food taken in the mouth is crushed into small particles by the grinding action of teeth.



DEGLUTITION is the process by which food is passed into the stomach from the oral cavity.


Throat
(Pharynx)
Upper
Esophageal
Sphincter (UES)
Trachea
Esophagus
Lower
Esophageal
Sphincter (LES)
Diaphragm
Stomach
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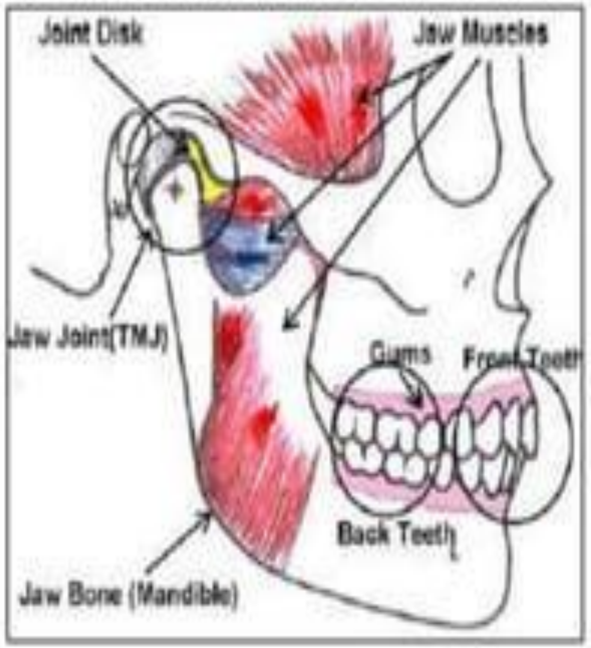
- Mastication consists of the coordinated functions of various parts of the oral cavity to prepare the food for swallowing and digestion.
- Large food particles are broken down and mixed with secretions of salivary glands to form a homogenized mass called **BOLUS**.
- This wetting and homogenizing action aids in swallowing and subsequent digestion.

- Teeth, lips, cheeks, tongue, palate, gingiva and periodontium, muscles of mastication and the temporomandibular joint, along with the lubricating and enzymatic action of saliva are involved in mastication.



IMPORTANCE OF MASTICATION:

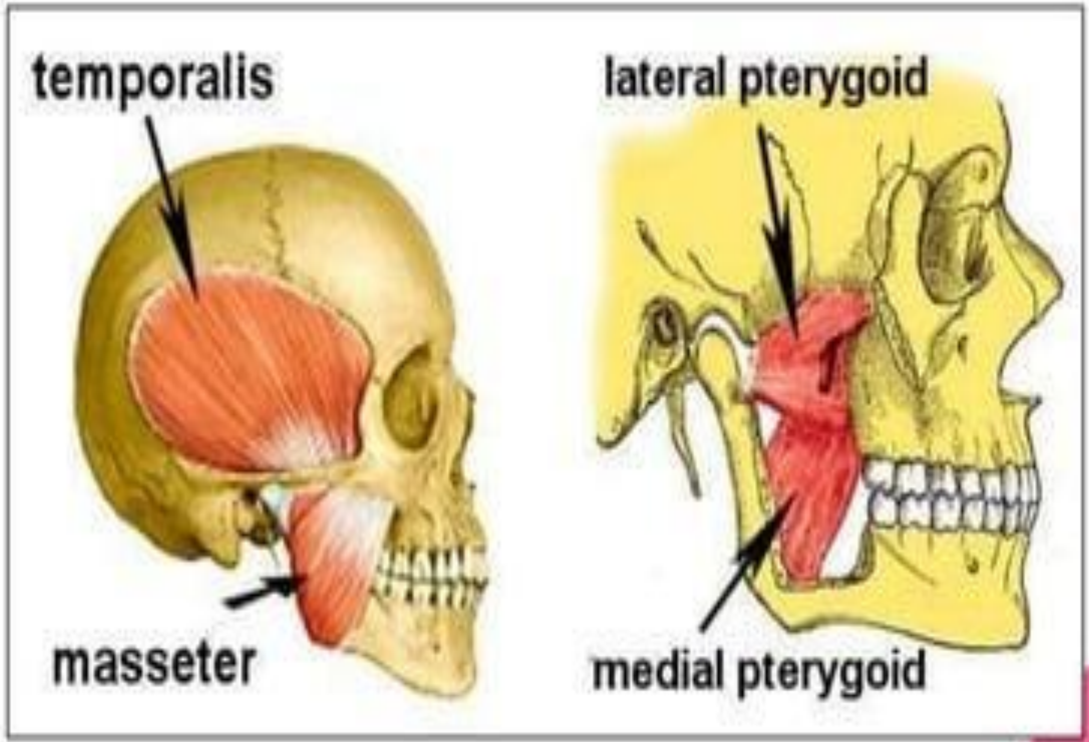
1. Large food particles can be digested, but they cause strong and often painful contractions of the oesophageal musculature.
2. Particles that are small tend to disperse in the absence of saliva, and this can make swallowing difficult.

3. Breakdown of food into smaller particles increase the surface area, resulting in effective digestion.

4. Mastication also helps in the removal of cellulose covering certain food particles.

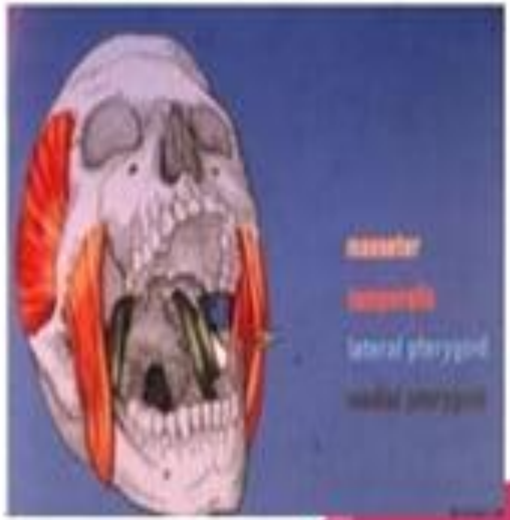
5. Saliva secretion is increased, and it lubricates the content of the oral cavity, aiding in easier swallowing.

MUSCLES OF MASTICATION



MUSCLES OF MASTICATION

These are 4 pairs of muscles that are attached to the mandible and are primarily responsible for elevating, protruding, retruding, or causing the mandible to move laterally.



- They develop from the first pharyngeal arch.
- Innervation- Trigeminal Nerve.
- Arterial supply- Maxillary Artery.

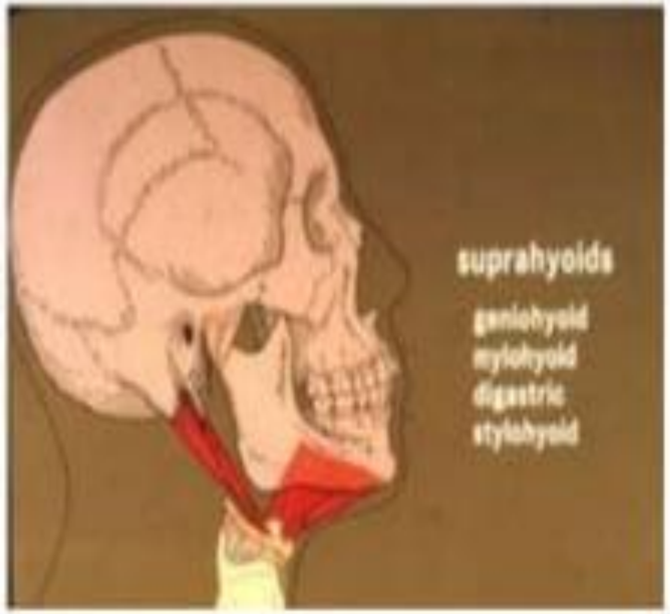
ACCESSORY MUSCLES OF MASTICATION

Anterior belly of Digastric

Geniohyoid

Mylohyoid

Buccinator



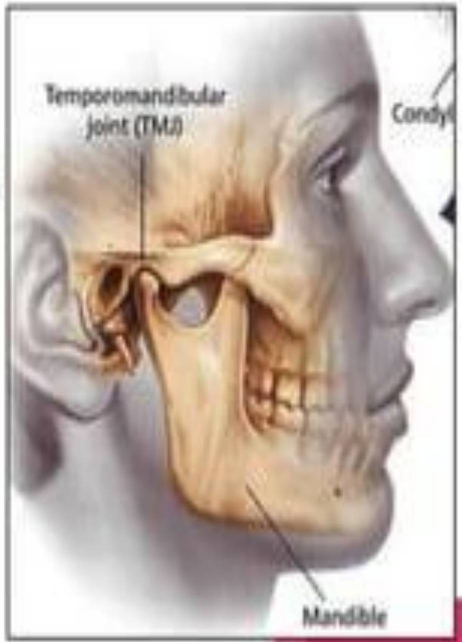
UNIQUE FEATURES OF MASTICATORY MUSCLES

- Have shorter contraction times than most other body muscles.
- Incorporate more of muscle spindles to monitor their activity.
- Do not have golgi tendon organs to monitor tension.
- Elevators are predominantly white fibrous which perform fast twitching.

- Do not fatigue easily.
- Psychological stress increases the activity of elevator muscles.
- Occlusal interferences can cause a hypertonic synchronous muscle activity.
- Closing movement also determined by the height of the teeth.

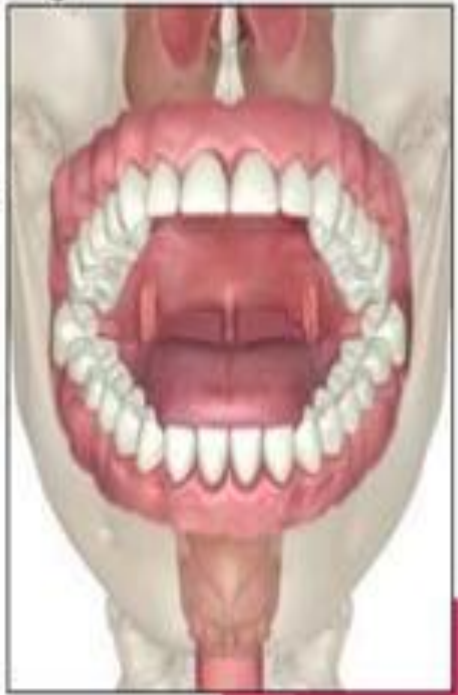
TEMPOROMANDIBULAR JOINT

- Temporomandibular Joint is a synovial joint that articulates between the mandible and the temporal bone.
- It connects the mandible to the skull and regulates its movement to perform important functions like speech and mastication.
- It is a bicondylar joint, located at either end of the mandible.



TONGUE

- The tongue plays an essential role in controlling the movement of food and formation of bolus.
- The food has to be positioned by the tongue in conjunction with the buccinator muscles between the occlusal surfaces of the teeth.
- The tongue transports solid and liquid food within the oral cavity.



SALIVA

- Saliva contain ZYMOGEN granules, which contain salivary enzymes that are discharged from the acinar cells into ducts.

PAROTID GLAND	SEROUS
SUBMANDIBULAR GLAND	MIXED
SUBLINGUAL GLAND	MUCOUS

- 1500 mL of saliva is secreted everyday.
- pH of resting gland is slightly <7; can go upto 8 during active secretions.

SALIVA contains-

- Digestive enzymes like lingual lipase and salivary α amylase.
- Mucins and glycoproteins- lubricate food, bind bacteria, protect the oral mucosa.
- Immunoglobulins, specifically secretory IgA
- Lysozymes that attack bacterial walls
- Lactoferrin- binds iron and is a bacteriostatic
- Proline rich proteins- protect the enamel and bind toxic tannins

- The digestive functions of saliva include moistening food and helping to create a food bolus. This lubricative function of saliva allows the food bolus to be passed easily from the mouth into the esophagus.
- Enzyme amylase, also called ptyalin, is capable of breaking down starch into simpler sugars. About 30% starch digestion takes place in the mouth cavity.
- Salivary lipase initiates fat digestion.

MASTICATORY SEQUENCE

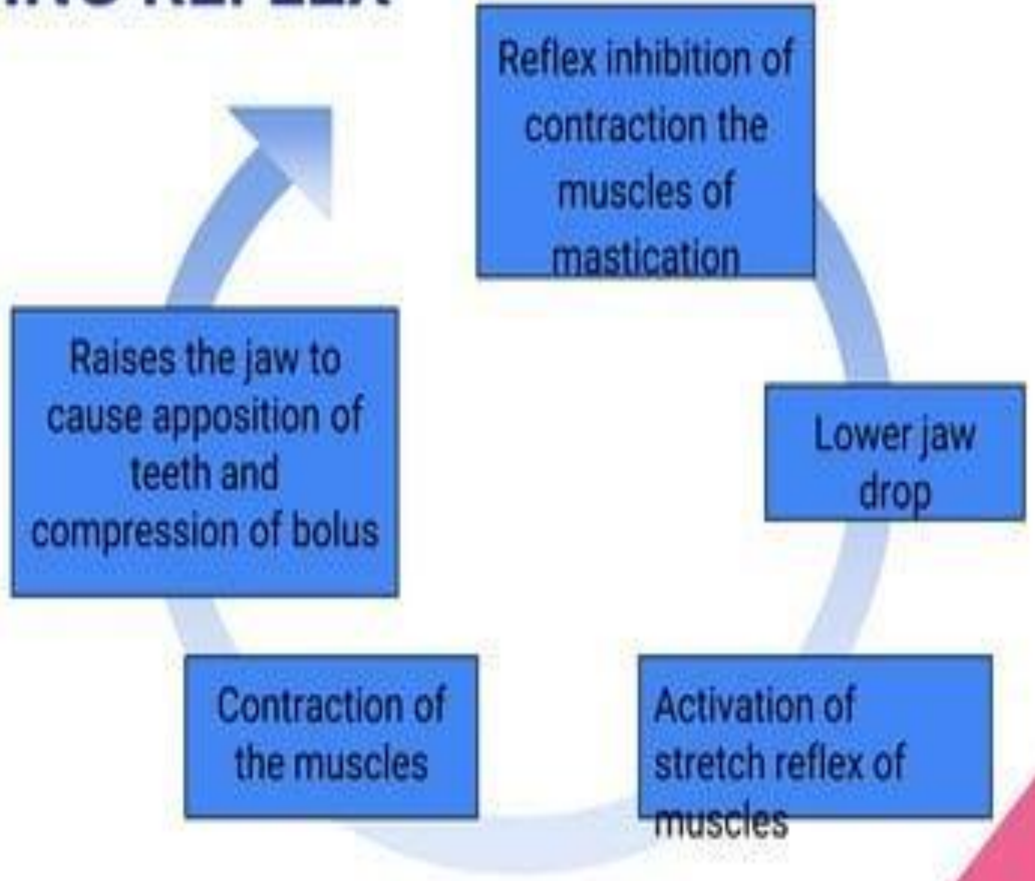
- Consists of many chewing cycles and extends from ingestion to swallowing.
- Can be divided into three consecutive periods-

1. PREPARATORY- Initial period where food is transported to the posterior teeth
2. REDUCTION- Intermediate period where food is broken down
3. PRE SWALLOWING- Final period where bolus is formed for swallowing.

CONTROL OF MASTICATION

- Motor and sensory nuclei contained in the brainstem play a pivotal role in the control of mastication.
- These movements require the integrated activity of muscles controlled by the Trigeminal, Facial and Hypoglossal nerves, and other motor nuclei in the brainstem.

CHEWING REFLEX



- The mastication process is controlled by a 'Central Pattern Generator' and masticatory muscles exhibit specific reflex patterns.
- Types of muscle reflexes of the masticatory apparatus-

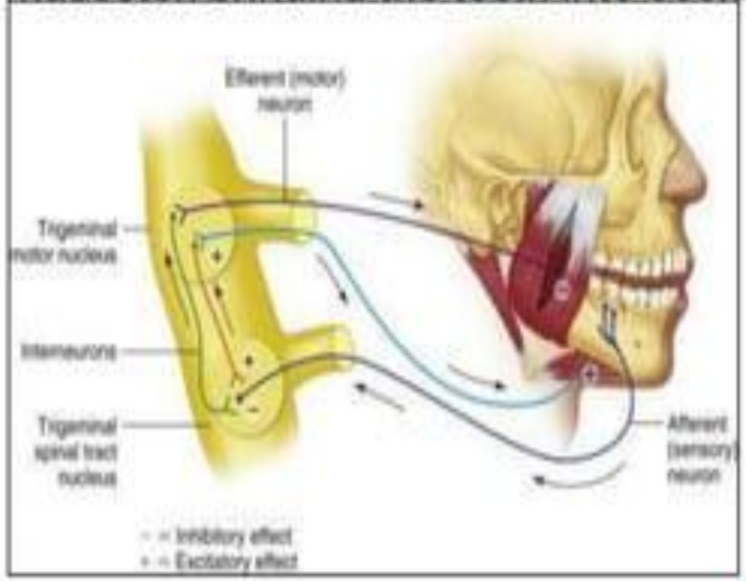
1. Jaw- closing reflex
2. Jaw- opening reflex
3. Unloading reflex

JAW- OPENING REFLEX-

- This reflex is initiated by mechanical stimulation of the periodontal ligament and mucosal mechanoreceptors.
- The result is excitation of jaw- opening muscles and inhibition of jaw- closing muscles.
- It is a polysynaptic reflex.
- There are numerous reflex pathways involved in this reflex.

JAW- CLOSING/ JAW- JERK REFLEX-

- The reflex activity of the jaw moving the teeth into occlusion.
- It is a monosynaptic reflex occuring due to stretching of muscle spindles in the masseter, bringing the teeth into occlusion.



UNLOADING REFLEX-

- It is a protective reflex that occurs during a heavy occlusal load.
- Mastication is stopped due to reflex inhibition of elevators and reflex excitement of depressors.
- Receptors in the periodontal ligament protect teeth from damage.